# Assignment 8 - Task List, Milestones, Timeline, and Effort Matrix

**Team Members:** Nate Heath, Elliot Warner, William Thomas

**Project Title:** Data-Day

**Goal Statement:** To build and open source a 64 x 32 pixel desktop display and the developer platform for creating applications for the device.

## Task List

1. Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module – Elliot Warner
2. Design the input-event interface for rotation and press in the Dial Button module – Elliot Warner
3. Develop dial rotation and button press detection when debouncing for the Dial Button module – Elliot Warner
4. Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module – Elliot Warner
5. Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module – Elliot Warner
6. Research operating-system and service requirements for application management on the Orange Pi module – William Thomas
7. Design the application lifecycle and state transitions for the Orange Pi module – William Thomas
8. Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module – William Thomas
9. Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module – William Thomas
10. Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module – William Thomas
11. Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module – Nate Heath
12. Design the framebuffer format for the Display module – Nate Heath
13. Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module – Nate Heath
14. Test edge coordinates, malformed frames, and refresh stability in the Display module – Nate Heath
15. Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device – Nate Heath

## Milestones

1. [COMPLETE] Team formation and basic project description – 9/9/26

   Report the team's members and a simple project overview for leading idea that includes the problem statement and proposed solution with a general explanation of the implementation. Verified via Canvas submission by Professor Zhan and via email by project mentor Professor Aurisano.

2. [COMPLETE] High-level system design – 10/2/26

   Create a high-level overview of the system and how it will work without going into in-depth implementation details. Include team member responsibility for different parts of the system. Verified via Canvas submission by Professor Zhan.

3. [COMPLETE] Detailed system design complete – 10/14/26

   Build out lower-level implementation details from high-level system design, including pseudocode for important algorithms and specific tools to use. Include possible edge cases and responses to edge cases. Verified via Canvas submission by Professor Zhan.

4. Requirements list – 10/30/26

   List of all project requirements along with technical specifications for the entire project's scope. Requirements are specific, testable, verifiable, etc. Verified via Canvas submission by Professor Zhan.

5. Presentation – 11/18/26

   Overview of problem statement and stakeholders, followed by proposed solution at varying levels of depth, as well as future work and goals. Mentions potential roadblocks and proposed workarounds. Oral presentation accompanied by a slideshow. Verified in person and via Canvas submission by Professor Zhan.

6. Design notebook – 12/2/26

   Complete documentation of progress up until point of submission. Covers overall project goals, covers design decisions made throughout the project's lifecycle, team member roles, etc. Includes tracking documentation (timeline, GANTT chart, etc.) of future work. Verified via Canvas submission by Professor Zhan.

7. Working basic prototype (no SDK) – 1/7/27

The Orange Pi can display customizable pixels on the LED display in a comprehensible manner. Driver work is completed to a level of functionality where coded pixels can be displayed on the LED display. Verified via demo in team meeting by project mentor Professor Aurisano.

8. Define SDK methods – 2/2/27

   Comprehensive list of SDK methods/endpoints with descriptions of what they do. Does not include implementation. Pseudocode included where useful. Verified via discussion and demos in team meeting by all team members.

9. Complete SDK – 2/14/27

   Comprehensive list of SDK methods/endpoints are filled in with implementations with code. Code is commented and simple example snippets are provided when necessary. Verified via discussion and demos in team meeting by all team members.

10. A sample program can run on machine – 3/7/27

    Verified via demo in team meeting by project mentor Professor Aurisano.

11. A user can upload a valid Python file – 3/15/27

    Verified via demo in team meeting by project mentor Professor Aurisano.

12. Full path works – 3/22/27

    Verified via demo in team meeting by project mentor Professor Aurisano.

13. Demo programs complete – 3/29/27

    Verified via demos in team meeting by project mentor Professor Aurisano.

14. Documentation complete – 4/5/27

    Verified via discussion in team meeting by all team members.

15. Complete demo – 4/15/27

    Verified via demo in team meeting by project mentor Professor Aurisano.

## Timeline

One row per Task List item with the task ID, statement, owner, start date, completion date, and dependencies, plus a row giving each milestone's planned completion date. Show your

buffer explicitly: 15 to 20 percent per task plus a protected block before each major milestone. A Gantt figure is welcome in addition, but the table is the graded item.

| Task ID | Statement | Owner | Start Date | Completion Date | Dependencies | Planned Completion Date | Buffer (Days) | Ri ID |
|---|---|---|---|---|---|---|---|---|
| T1 | Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module | Elliot Warner | 12/1/26 | N/A | None | 12/5/26 | 1 | R5 |
| T2 | Design the input-event interface for rotation and press in the Dial Button module | Elliot Warner | 12/5/26 | N/A | None | 12/10/26 | 1 | |
| T3 | Develop dial rotation and button press detection when debouncing for the Dial Button module | Elliot Warner | 12/10/26 | N/A | T1, T2 | 12/15/26 | 1 | |
| T4 | Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module | Elliot Warner | 12/15/26 | N/A | T3 | 12/20/26 | 1 | |
| T5 | Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module | Elliot Warner | 1/4/27 | N/A | T1, T2 | 2/9/27 | 7 | |
| T6 | Research operating-system and service requirements for application | Will Thomas | 11/20/26 | N/A | None | 11/24/26 | 1 | R4 |

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| | management on the Orange Pi module | | | | | | | |
| T7 | Design the application lifecycle and state transitions for the Orange Pi module | Will Thomas | 12/5/26 | N/A | T6 | 12/15/26 | 2 | R4 |
| T8 | Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module | Will Thomas | 2/8/27 | N/A | T7 | 2/18/27 | 2 | R4 |
| T9 | Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module | Will Thomas | 2/18/27 | N/A | T8 | 2/26/27 | 2 | R4 |
| T10 | Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module | Will Thomas | 2/26/27 | N/A | T9 | 3/5/27 | 1 | R4 |
| T11 | Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module | Nate Heath | 11/25/26 | N/A | T6, T7 | 11/30/26 | 1 | R2 |
| T12 | Design the framebuffer format for the Display module | Nate Heath | 11/30/26 | N/A | T11 | 12/7/26 | 1 | R2 |
| T13 | Develop framebuffer | Nate Heath | 12/7/26 | N/A | T12 | 12/16/26 | 2 | R1 |

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| | operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module | | | | | | | |
| T14 | Test edge coordinates, malformed frames, and refresh stability in the Display module | Nate Heath | 12/16/26 | N/A | T13 | 12/20/26 | 1 | R1 |
| T15 | Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device | Nate Heath | 3/5/27 | N/A | T4, T14 | 3/15/27 | 2 | R3 |

Note: Tasks may be blocked by tasks that have blocks so there may be more blocking areas than listed.

## Effort Matrix

| Task ID | Task | Primary Owner | Nate (hrs) | Elliot (hrs) | William (hrs) |
|---|---|---|---|---|---|
| T1 | Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module | Elliot Warner | 2 | 5 | 2 |
| T2 | Design the input-event interface for rotation and press in the Dial Button module | Elliot Warner | 3 | 10 | 2 |
| T3 | Develop dial rotation and button press | Elliot Warner | 2 | 10 | 3 |

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| | detection when debouncing for the Dial Button module | | | | |
| T4 | Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module | Elliot Warner | 2 | 12 | 3 |
| T5 | Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module | Elliot Warner | 1 | 4 | 1 |
| T6 | Research operating-system and service requirements for application management on the Orange Pi module | William Thomas | 2 | 2 | 10 |
| T7 | Design the application lifecycle and state transitions for the Orange Pi module | William Thomas | 3 | 3 | 12 |
| T8 | Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module | William Thomas | 5 | 5 | 26 |
| T9 | Validate package structure, metadata, file-size limits, and error | William Thomas | 2 | 2 | 12 |

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| | responses in the Orange Pi module | | | | |
| T10 | Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module | William Thomas | 3 | 3 | 18 |
| T11 | Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module | Nate Heath | 10 | 3 | 2 |
| T12 | Design the framebuffer format for the Display module | Nate Heath | 12 | 3 | 2 |
| T13 | Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module | Nate Heath | 22 | 3 | 2 |
| T14 | Test edge coordinates, malformed frames, and refresh stability in the Display module | Nate Heath | 12 | 3 | 2 |
| T15 | Integrate the Dial Button, Orange Pi, and Display modules and test application and | Nate Heath | 14 | 3 | 2 |

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| | selection on the complete device | | | | |
| **TOTALS** | | | 95 hrs (34.5%) | 71 hrs (26.8%) | 99 hrs (38.7) |

## Work Breakdown Structure

### Phase 1: Hardware and Low-Level Drivers

1.1) Input Hardware Investigation and Wiring (T1)

**Task Statement**: Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module.
**Primary Owner**: Elliot Warner

1.2) Display Driver Compatibility and Refresh Constraints (T11)

**Task Statement**: Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module.
**Primary Owner**: Nate Heath

1.3) Framebuffer Format and Memory Allocation (T12)

**Task Statement**: Design the framebuffer format for the Display module.
**Primary Owner**: Nate Heath

1.4) Framebuffer Operations and Text Rendering (T13)

**Task Statement**: Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module.
**Primary Owner**: Nate Heath

### Phase 2: OS and Core Application Management

2.1) Operating System & Service Requirements (T6)

**Task Statement**: Research operating-system and service requirements for application management on the Orange Pi module.
**Primary Owner**: William Thomas

**2**.2) Application Lifecycle and State Transitions (T7)

**Task Statement**: Design the application lifecycle and state transitions for the Orange Pi module.
**Primary Owner**: William Thomas

**2**.3) Application Runtime Management (T8)

**Task Statement**: Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module.
**Primary Owner**: William Thomas

**2**.4) Local Upload and Management Endpoints (T10)

**Task Statement**: Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module.
**Primary Owner**: William Thomas

## Phase 3: User Interface and Event Handling

**3.1**) Input-Event Interface Design (T2)

**Task Statement**: Design the input-event interface for rotation and press in the Dial Button module.
**Primary Owner**: Elliot Warner

**3.2**) Debouncing and Rotation Signal Handling (T3)

**Task Statement**: Develop dial rotation and button press detection when debouncing for the Dial Button module.
**Primary Owner**: Elliot Warner

## Phase 4: Testing and Package Validation

**4.1**) Dial Input Boundary & Rapid Input Testing (T4)

**Task Statement**: Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module.
**Primary Owner**: Elliot Warner

**4.2**) Package Structure & Size Limit Validation (T9)

**Task Statement**: Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module.
**Primary Owner**: William Thomas

**4.3**) Display Edge & Frame Stability Testing (T14)

**Task Statement**: Test edge coordinates, malformed frames, and refresh stability in the Display module.
**Primary Owner**: Nate Heath

Phase 5: Integration and Documentation

**5.1**) Module Documentation & Setup Guides (T5)

**Task Statement**: Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module.
**Primary Owner**: Elliot Warner

**5.2**) Full System Integration & Device Testing (T15)

**Task Statement**: Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device.
**Primary Owner**: Nate Heath

## Risk Register

| ID | Risk | Trigger | Mitigation | Owner | Likelihood | Impact | Strategy | Category | Taks |
|---|---|---|---|---|---|---|---|---|---|
| R1 | Display work that slips past 12/20 can't be recovered over winter break | T13 not complete by 12/16 | All three members work on the display build in parallel and share findings so it's done before 12/20 | Nate Heath | Medium | High | Reduce | Team | T13, T14 |
| R2 | Display driver/framebuffer work runs long | Any of T11–T14 misses its completion date | Demo a simpler version of the display output at M7 if the full framebuffer work | Elliot Warner | Low | Medium | Accept | Technical | T11-T14 |

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| | | | isn't ready | | | | | | |
| R 3 | Integration overruns its protected block | T15 not complete by 3/15 | Cut a feature from the Full path works demo so the core path is shown on 3/22 | Nate Heath | Low | Low | Accept | Technical | T15 |
| R 4 | Critical path rests on one member | T6 not complete by 11/24 | William cross-trains Nate and Elliot on the Orange Pi module by 1/14/27 so either can pick up T8–T10 | Elliot Warner | Low | High | Reduce | Team | T6 - T10 |
| R 5 | Hardware not on hand or incompatible | Hardware not operational by 12/1/26 | All hardware is already on hand; if a part breaks, order a replacement immediately | Nate Heath | Low | High | Reduce | External | T1, T11 |